Final Project Overview

Financial Data Science II, Fall 2020

Spoofing Example

I OWN GOOGLE STOCK AND WANT TO SELL AS HIGH AS POSSIBLE

I have an offer of 100 lots at 576.58

ASK	#
576.86	8
576.85	22
576.85	250
576.80	100
576.74	200
576.73	120
576.67	100
576.58	100

#	BID
300	576.42
75	576.40
500	576.40
10	576.37
10	576.36
8	576.26
2	576.26
30	576.25

I also add a bid of 20,000 lots at 576.42, to look like demand

ASK	#
576.86	8
576.85	22
576.85	250
576.80	100
576.74	200
576.73	120
576.67	100
576.58	100

#	BID
20,300	576.42
75	576.40
500	576.40
10	576.37
10	576.36
8	576.26
2	576.26
30	576.25

Scared, someone else lifts the offer and buys my stock at 576.58

ASK	#
576.86	8
576.85	22
576.85	250
576.80	100
576.74	200
576.73	120
576.67	100
576.58	20

#	BID
220	576.42
75	576.40
500	576.40
10	576.37
10	576.36
8	576.26
2	576.26
30	576.25

BOOM!

Market trades up at 576.58, and I sell 80 lots of Google Stock

This is called “Spoofing”

<u>i.e. Entering orders with no intent to trade</u>

It is highly illegal, and leads to huge fines for financial institutions!

Tower Research Hit with Record $67 Million Spoofing Fine

Tower made $32.6 million in profit from spoofing activity carried out by three former traders.

U.S. CFTC to fine UBS, Deutsche Bank, HSBC for spoofing, manipulation: sources

Merrill Lynch fined $25 million for multi-year spoofing scheme

Merrill Lynch Commodities admitted the illegal activity following an investigation by the US Department of Justice and the CFTC.

October 1, 2019

CFTC Orders Two Trading Firms, Bank to Pay a Total of $3 Million for Spoofing

Hackathon Challenge

CONTEXT:

Spoofing is the activity of entering trading orders with no intention to actually trade; to manipulate the market price. This is often done to benefit a traders current positions and increase their PnL.

This is a BIG PROBLEM. It carries financial and reputational risk, and is strictly prohibited by all major financial regulators.

Nevertheless, companies like Citi execute millions of trade orders every day, and it is impossible to track, observe and analyze every single trading order that is made.

GOAL:

To systematically describe, identify trading activity which has no desire to actually trade (i.e. spoofing), that can be applied to huge data sets of trading activity.

The Data

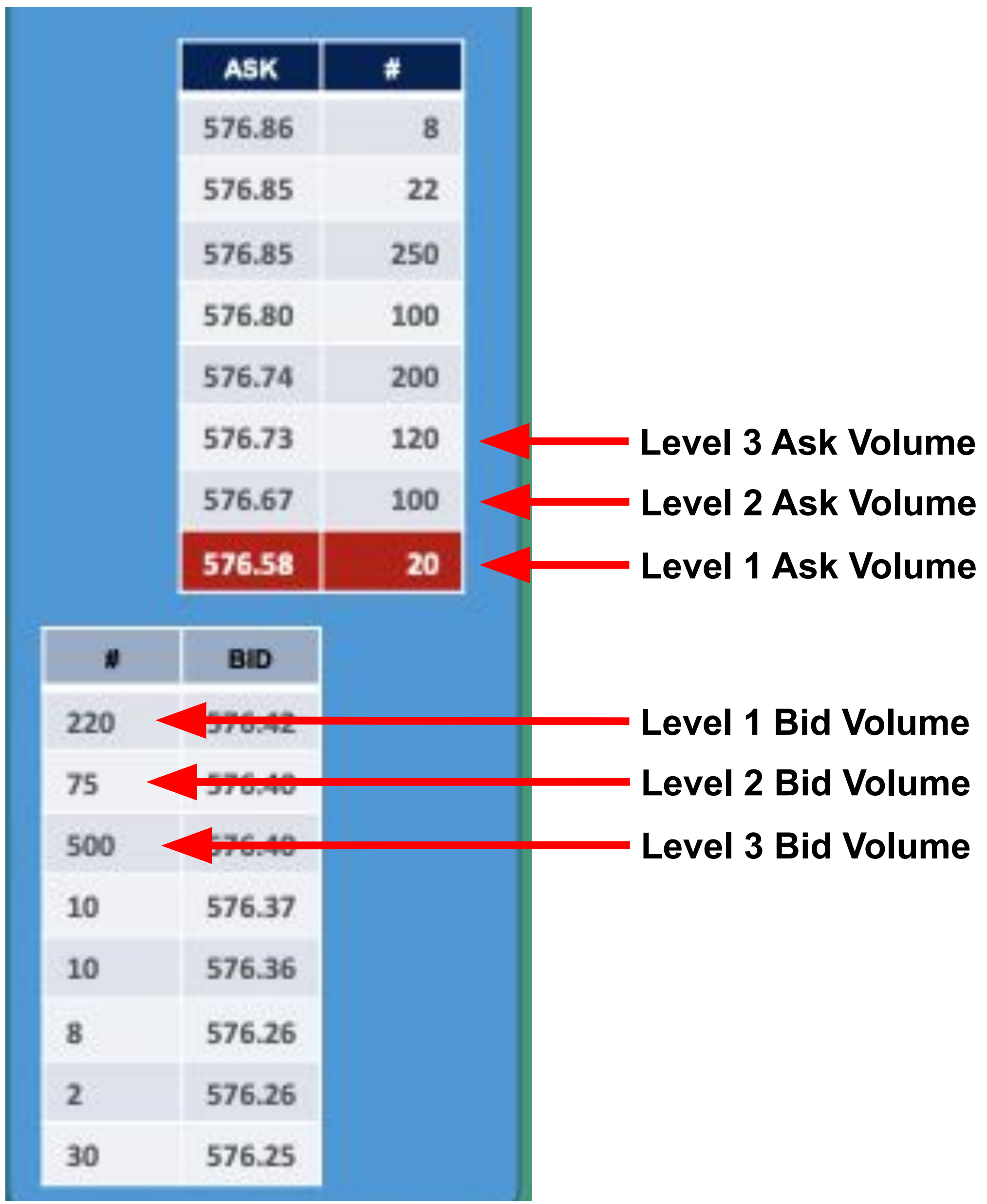
You will be provided with a sample of labelled **cancellation** orders from an order book

Ten Original Variables:

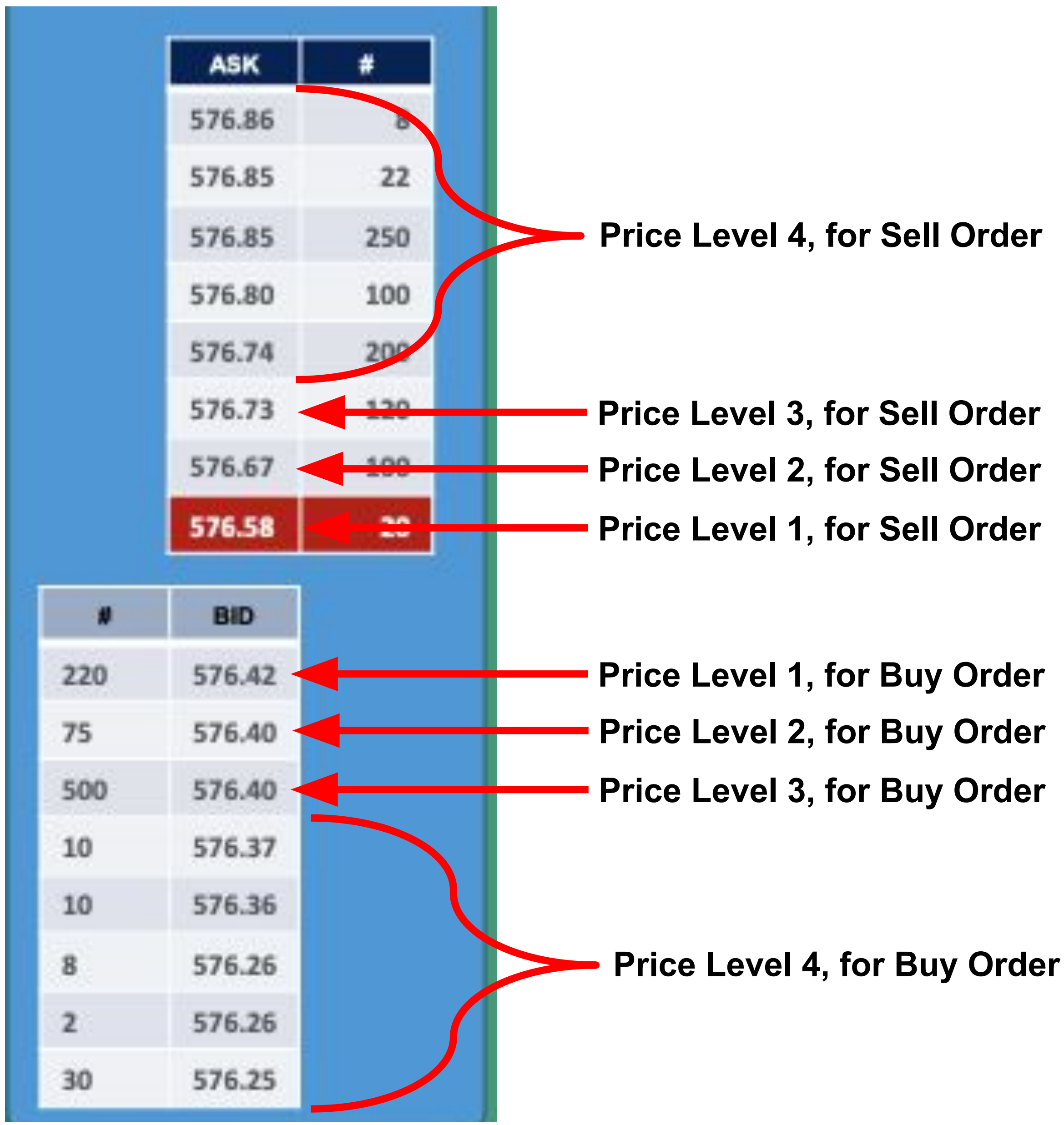
1. **spoofing** indicator
2. **size** of order*
3. **type** of order - buy or sell
4. **price** of order - at what level in the book is the order?*
5. **volume of book** at three lowest ask prices*
6. **volume of book** at three highest bid prices*

** These variables are binned in order to make the differing price and volume levels across stocks comparable*

The Data


ASK
#
576.86
8
576.85
22
576.85
250
576.80
100
576.74
200
576.73
120
576.67
100
576.58
20
Level 3 Ask Volume
Level 2 Ask Volume
Level 1 Ask Volume
#
BID
220
75
500
10
576.37
10
576.36
8
576.26
2
576.26
30
576.25
Level 1 Bid Volume
Level 2 Bid Volume
Level 3 Bid Volume

The Data


ASK
#
576.86
8
576.85
22
576.85
250
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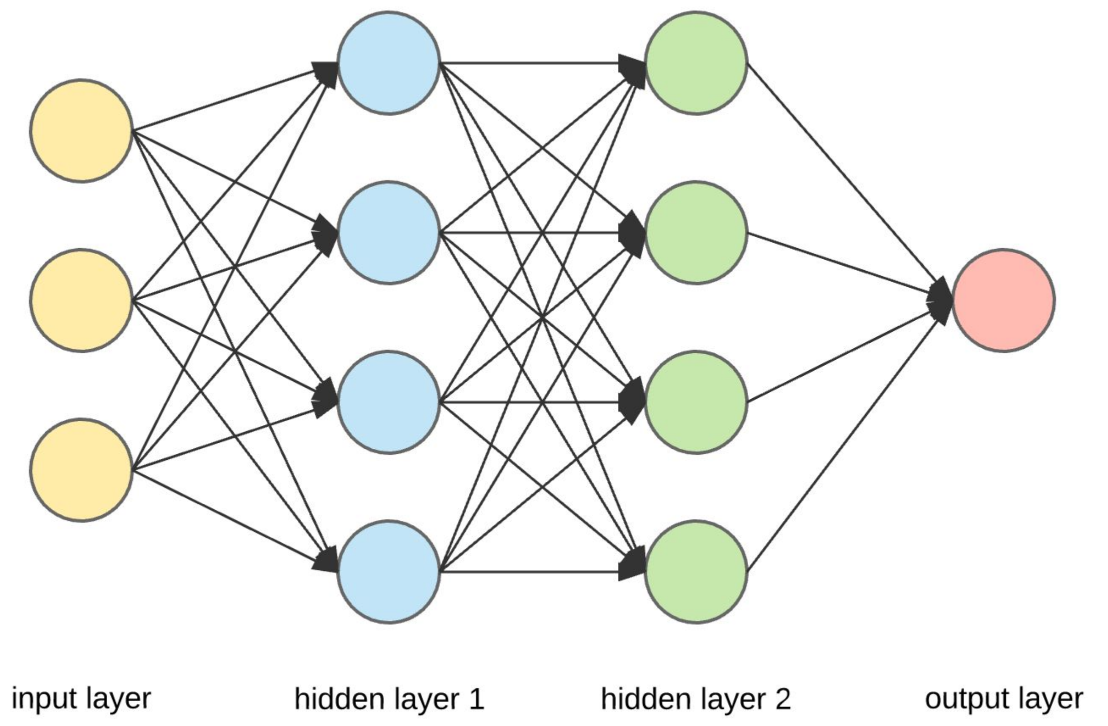
576.74
200
576.73
120
576.67
100
576.58
20
#
BID
220
576.42
75
576.40
500
576.40
10
576.37
10
576.36
8
576.26
2
576.26
30
576.25
Price Level 4, for Sell Order
Price Level 3, for Sell Order
Price Level 2, for Sell Order
Price Level 1, for Sell Order
Price Level 1, for Buy Order
Price Level 2, for Buy Order
Price Level 3, for Buy Order
Price Level 4, for Buy Order

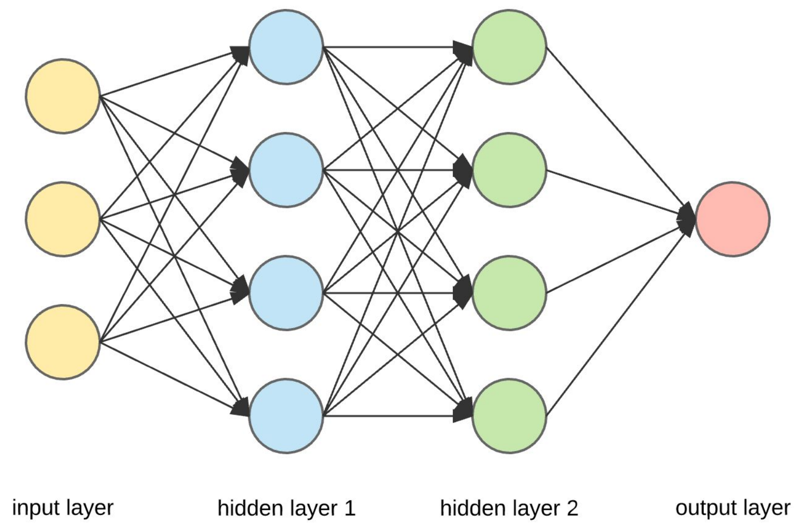
Classification

You are **not** being asked in this challenge to build a classifier.

I trained a neural network to separate spoofing from non-spoofing based on the available features



Classification



The neural network fit here has nine input features, and ten nodes in the final hidden layer.

On a modest test set, precision and recall were both about 92% for both classes.

Classification

But: There is a concern with interpretability of the results

“Explainable Machine Learning/AI”

The challenge to you is to

1. characterize the features used by the neural network
2. relate the features used by the neural network to the original variables

You will be provided with the ten values from the final hidden layer of the network.

Guidelines

Put yourself in the position of answering the question: How is the neural network utilizing the original data to make the predictions?

Imagine that you are trying to explain the results to an individual not familiar with machine learning methods.

I do **not** want a tutorial on neural networks or an attempt to “reverse engineer” my neural network. This project does not require any knowledge of how a neural network functions.

Explore the data, and how the variables are related.

Guidelines

What do we want to see?

1. Visualization (with clear axis labels, etc.)
2. Use of smoothing to highlight relationships
3. Use of dimension reduction and clustering
4. Clear explanations (but not excessively wordy)

Logistics

The template notebook will be available at 5:00 PM EST on Thursday, December 10

All work will be done on Google Colab

Make a copy of the template notebook

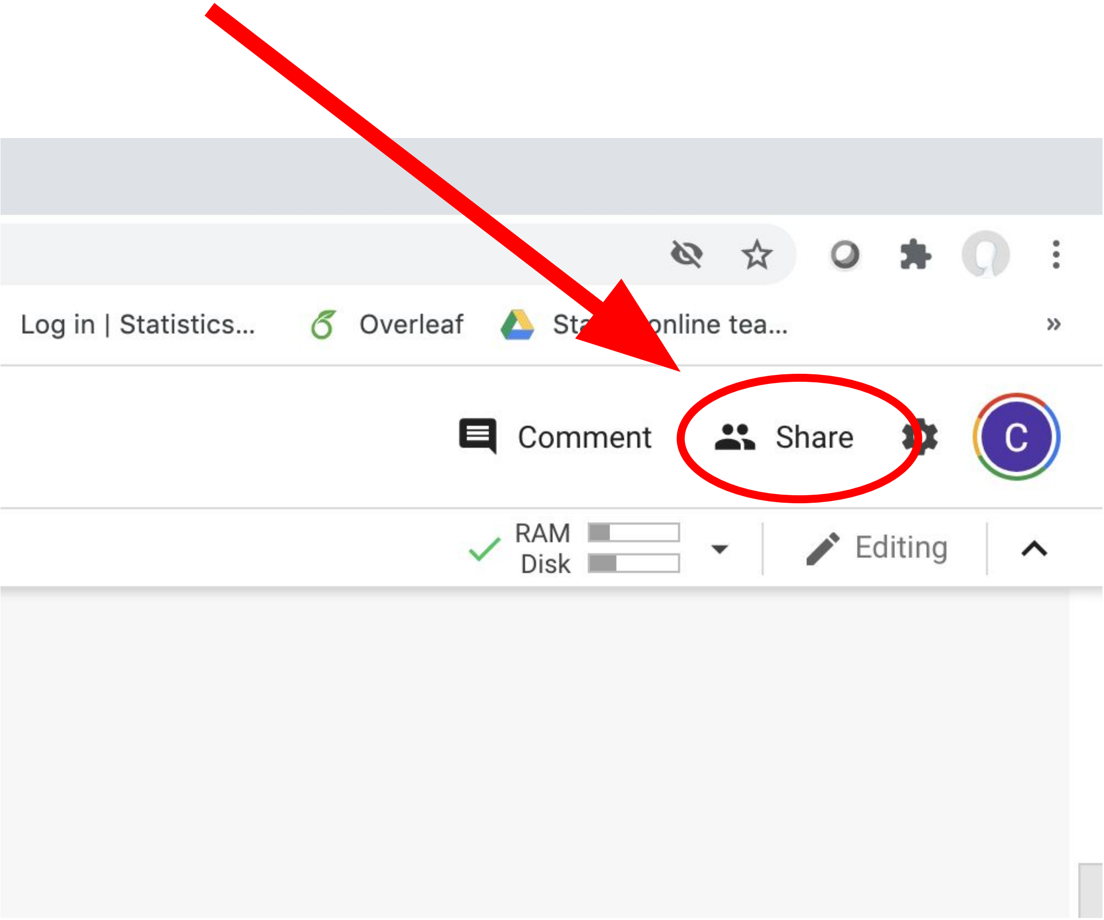
Name the notebook using your Andrew ID

Share your notebook with me **immediately**

You can spend eight hours working on the project. All work must be done in the notebook. The notebook is the only thing which will be assessed and graded.

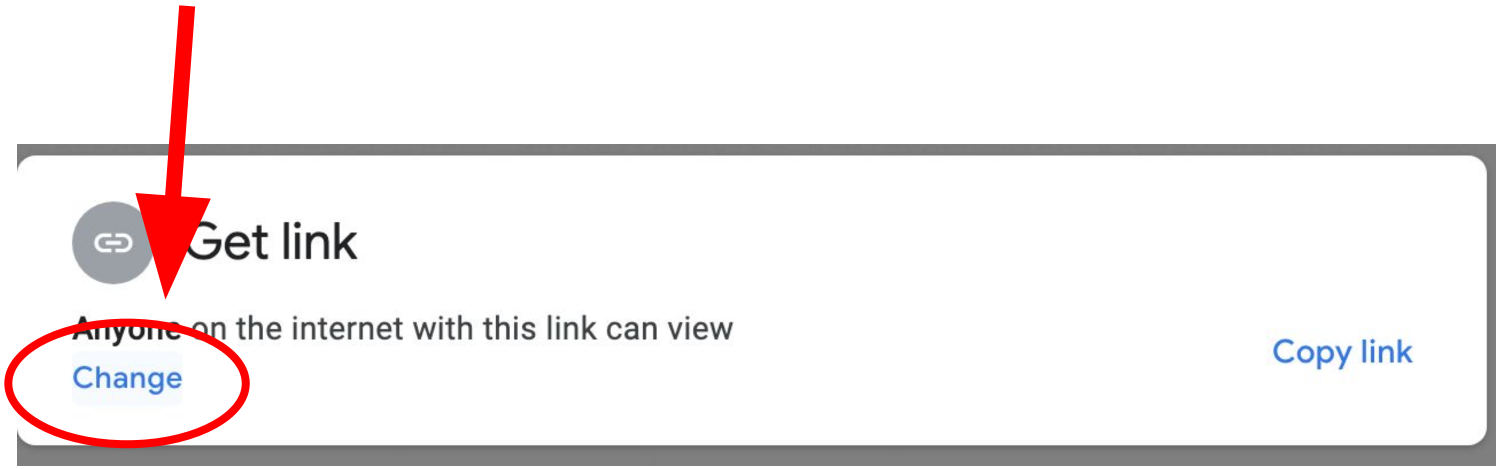
How to Share your Notebook

1. Click on “Share” in the upper right



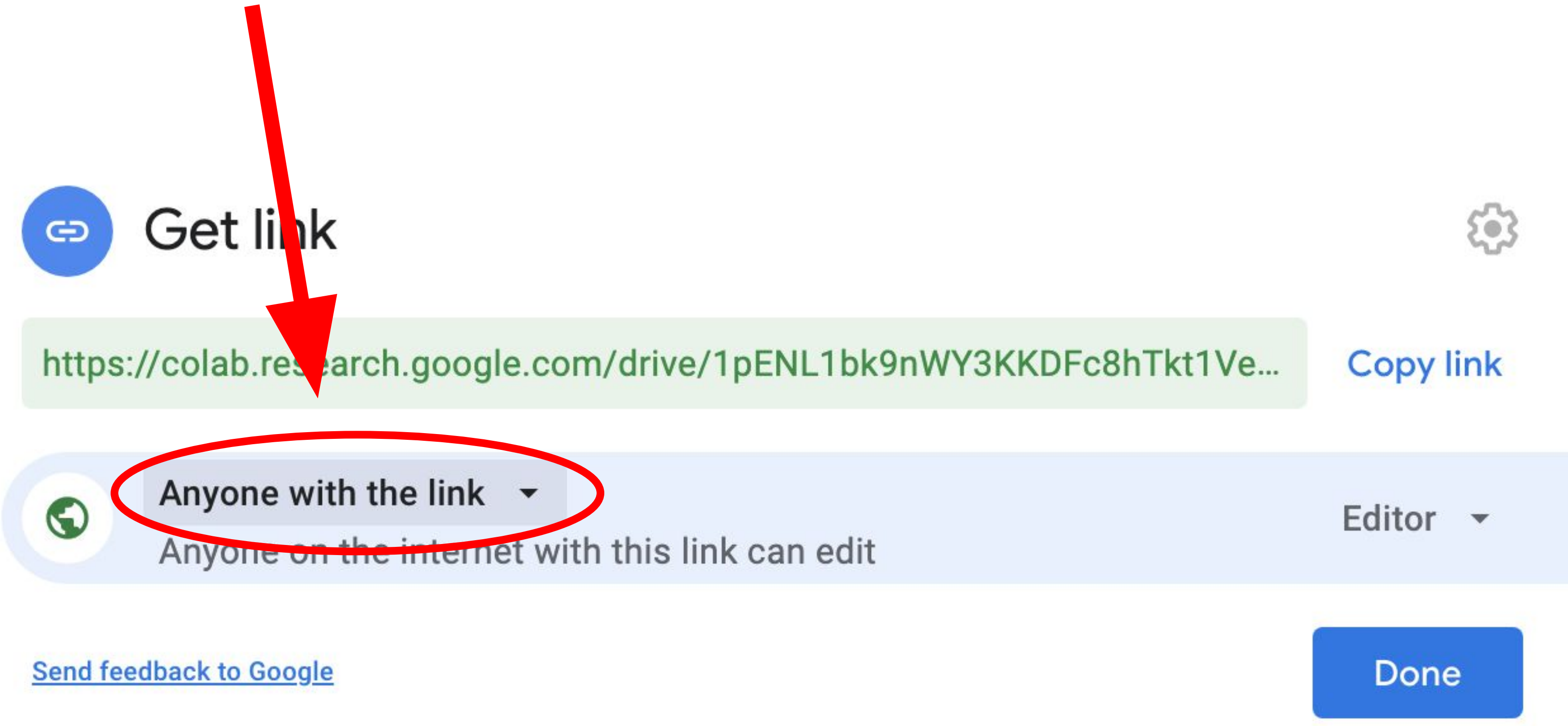
How to Share your Notebook

2. Click on “Change” in the lower left



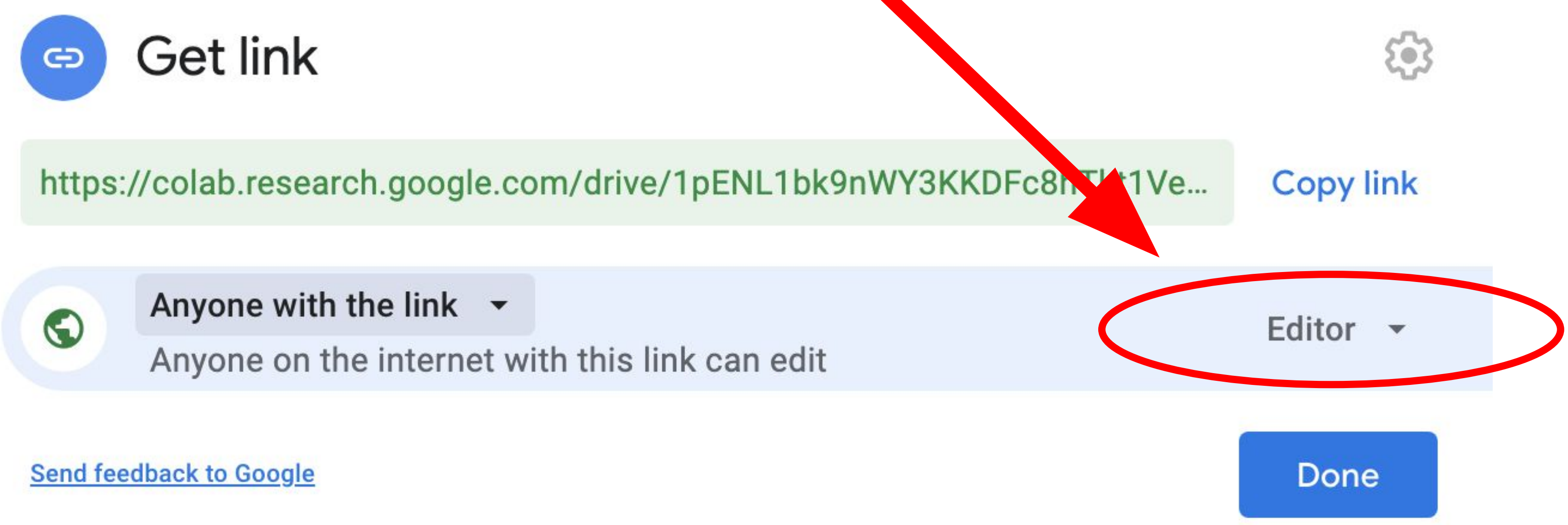
How to Share your Notebook

3. Make sure the option “Anyone with the link” is chosen



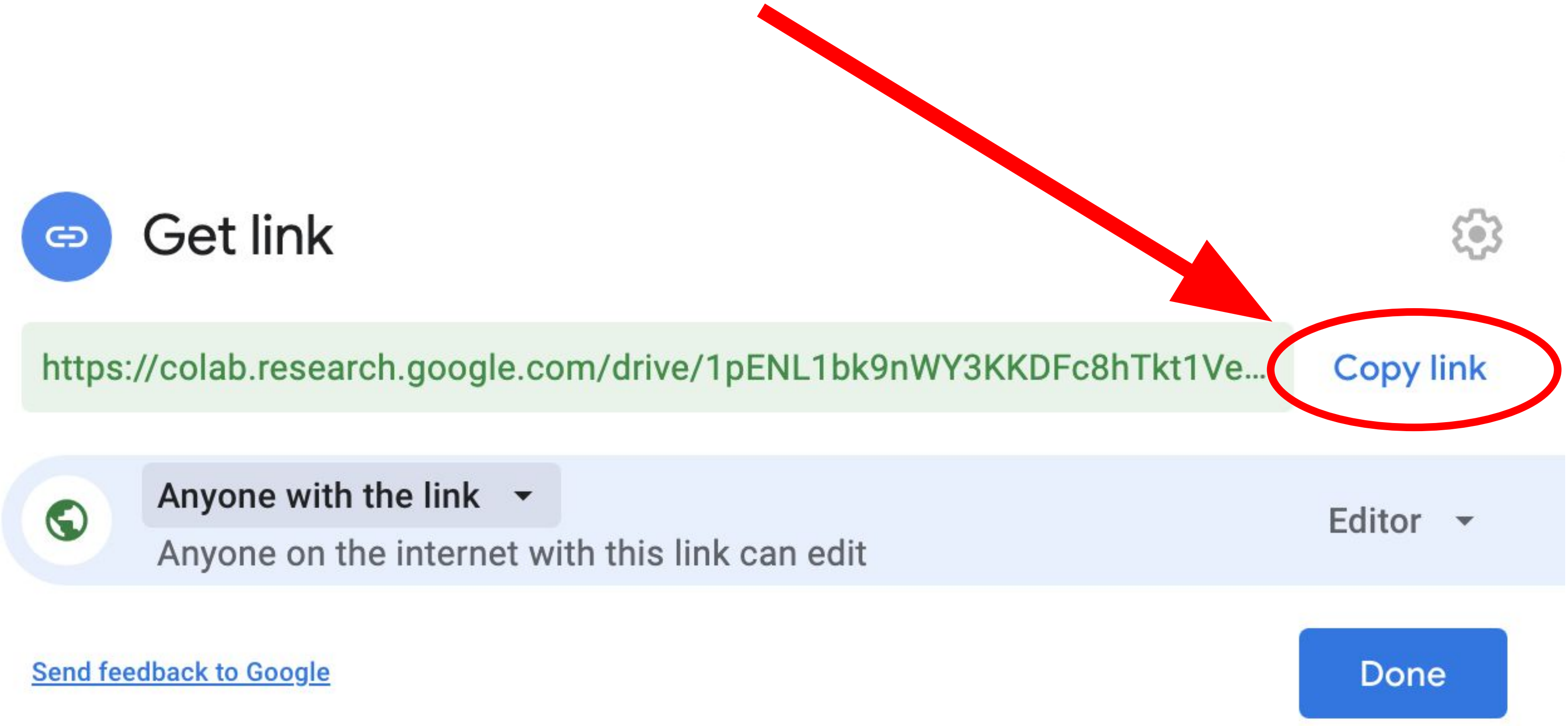
How to Share your Notebook

4. Make sure the option “Editor” is chosen



How to Share your Notebook

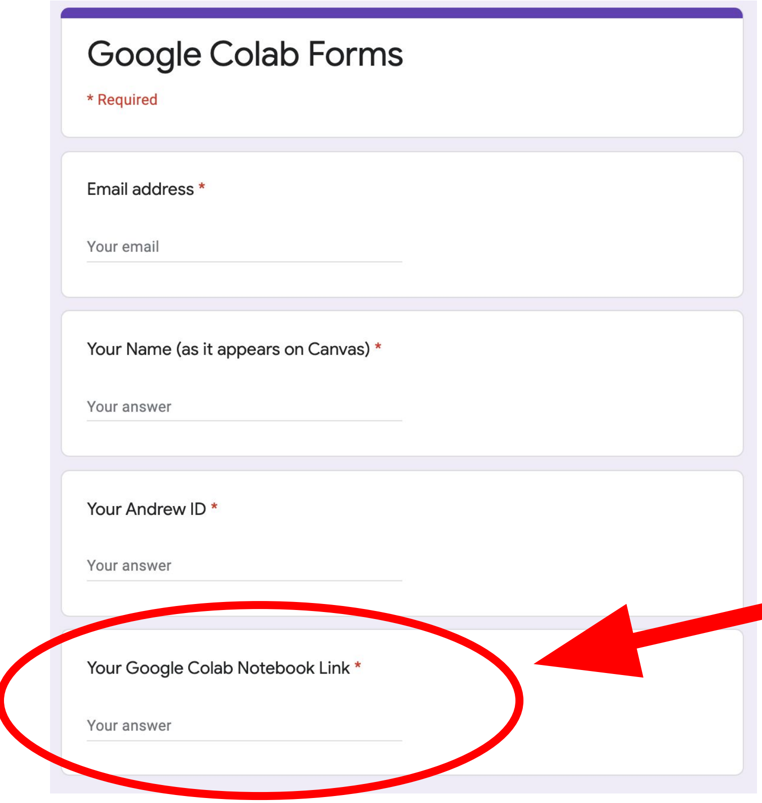
5. Click on “Copy Link” and then click “Done”



How to Share your Notebook

6. Go to the Google Form, and give your name and Andrew ID, and paste the link.

https://forms.gle/Kj5SKrqY8jwcc9y79



Your notebook link (copied in the previous step) goes here.

Test Notebook

Access this notebook, and try making a copy of it:

https://colab.research.google.com/drive/18-ruuPfQNpp1kKkd7xQu84v1W_sBBmWP?usp=sharing

Take some time before tomorrow to make sure you can access the notebook, and to learn how it works.

It is much like Jupyter notebooks.